■# PAPER_177: FluidSolver Navier-Stokes + UQFF Coupling — Quasar Jet Dynamics

Whitepaper §2.4-I | Thread 381a8fe7 | Session 48

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Abstract

The CoAnQi codebase implements a 2D incompressible Navier-Stokes solver (FluidSolver) on a 32×32 grid, driven by the UQFF resonance gravity as an external body force. This enables simulation of quasar jet dynamics as a coupled UQFF-fluid system. This paper documents the solver algorithm, boundary conditions, UQFF coupling interface, and jet injection mechanism.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?[1], [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa\cdot[SSq]\cdot\frac{GM}{r^2}, \quad \kappa = 5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57,\; \beta_i = 0.61$$

1. Configuration Parameters

```
class FluidSolver {
const int N = 32; // Grid resolution (32×32)
const double dt = 0.1; // Timestep [s]
const double visc = 0.0001; // Kinematic viscosity [m²/s]
double force_jet = 10.0; // Jet injection force [N/m²]
// State arrays: ux[N+2][N+2], uy[N+2][N+2], p[N+2][N+2], rho[N+2][N+2]
// Previous: ux_prev, uy_prev, p_prev
};
```

---

2. Solver Algorithm

The solver follows the Stam (1999) stable fluids method with four phases:

Phase 1 — Add Source

```
ux[i][j] += dt × uqff_g (external gravity from compute_resonance_MUGE)
```

Phase 2 — Diffuse

```
diffuse(N, 1, ux_prev, ux, visc, dt):
a = dt × visc × N²
for k in 0..19: // 20 Gauss-Seidel iterations
for i,j in 1..N:
ux[i][j] = (ux_prev[i][j] + a*(ux[i-1][j]+ux[i+1][j]+ux[i][j-1]+ux[i][j+1]))
/ (1 + 4a)
```

```
set_bnd(N, 1, ux)
```

Phase 3 — Project (Pressure solve / Helmholtz decomposition)

```
project(N, ux, uy, p, div):
h = 1.0 / N
// Compute divergence
div[i][j] = -0.5h*(ux[i+1][j]-ux[i-1][j] + uy[i][j+1]-uy[i][j-1])
p[i][j] = 0
// Pressure Poisson (20 iterations)
for k in 0..19:
p[i][j] = (div[i][j]+p[i-1][j]+p[i+1][j]+p[i][j-1]+p[i][j+1]) / 4
// Subtract pressure gradient
ux[i][j] -= 0.5*(p[i+1][j]-p[i-1][j]) / h
uy[i][j] -= 0.5*(p[i][j+1]-p[i][j-1]) / h
set_bnd(N, 1, ux); set_bnd(N, 2, uy)
```

Phase 4 — Advect (Semi-Lagrangian backtracing)

```
advect(N, b, d, d0, ux, uy, dt):
dt0 = dt × N
for i,j in 1..N:
x = i - dt0*ux[i][j] // backtrack position
y = j - dt0*uy[i][j]
// Clamp and bilinear interpolate d0 at (x,y)
d[i][j] = bilinear_interp(d0, x, y)
set_bnd(N, b, d)
```

---

3. Boundary Conditions (set_bnd)

```
b=1 (x-velocity): ux[0][j] = -ux[1][j] (no-slip left/right walls)
b=2 (y-velocity): uy[i][0] = -uy[i][1] (no-slip top/bottom walls)
b=0 (scalar): zero-gradient Neumann on all walls
Corner cells: average of two adjacent boundary cells
```

---

4. UQFF Coupling Interface

```
void step(double uqff_g) {
// uqff_g = compute_resonance_MUGE(muge_system)
// Apply UQFF as external body force
for i in 1..N:
for j in 1..N:
ux[i][j] += dt × uqff_g; // gravity drives x-velocity
std::swap(ux_prev, ux);
diffuse(N, 1, ux_prev, ux, visc, dt);
diffuse(N, 2, uy_prev, uy, visc, dt);
project(N, ux, uy, p, div);
std::swap(ux_prev, ux); std::swap(uy_prev, uy);
advect(N, 1, ux, ux_prev, ux_prev, uy_prev, dt);
advect(N, 2, uy, uy_prev, ux_prev, uy_prev, dt);
project(N, ux, uy, p, div);
}
```

---

5. Jet Force Injection

```
void add_jet_force() {
// Injects a transverse jet force at the domain midpoint
for i in N/4..3N/4: // Inject across central 50% of grid
uy[i][N/2] += force_jet; // force_jet = 10.0 N/m²
}
```

This models quasar jet ejection as a sustained transverse force at the equatorial plane, consistent with the UQFF interpretation of SCm expulsion igniting along the Ug4 galactic axis.

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6. Velocity Field Visualisation

```
void print_velocity_field() {
// ASCII: '#'=|v|>1, '+'=|v|>0.5, '.'=|v|>0.1, ' '=|v|=0.1
for i in 1..N:
for j in 1..N:
mag = sqrt(ux[i][j]^2 + uy[i][j]^2)
print( (mag>1)?'#': (mag>0.5)?'+': (mag>0.1)?'.':' ' )
}
```

---

7. integratefluidsfor_muge() — System-Level Integration

```
void simulate_fluids_for_muge(MUGESystem& sys, int N_steps=100) {
FluidSolver fs;
for int step=0; step<N_steps; step++:
double g = compute_resonance_MUGE(sys); // UQFF gravity
fs.add_jet_force();
fs.step(g);
fs.print_velocity_field();
}
```

This function is called from `populate_simulation_entities()` for each system in the 7-system canonical set, coupling fluid dynamics to UQFF.

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8. Physical Interpretation

Solver Component	Physical Analogue
ux, uy velocity field	Quasar jet plasma velocity
uqff*g (resonance*MUGE)	UQFF gravitational drive
visc = 0.0001	Magnetised plasma viscosity
add*jet*force	SCm expulsion ignition event
project (pressure)	Magnetohydrodynamic pressure balance
set_bnd (no-slip)	Accretion disk boundary

---

9. References

FluidSolver.h/cpp (thread 381a8fe7)

Stam, J. (1999) "Stable Fluids" — SIGGRAPH proceedings

PAPER*174 (resonance*MUGE provides uqff_g)

PAPER_176 (SCm expulsion as jet trigger)

PAPER_178 (3D simulation entities that host fluid simulations)